ADITHYA CHERIAN ABRAHAM

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RESEARCH STATEMENT

Final-year MCA student at Amrita Vishwa Vidyapeetham focused on deep learning architectures, representation learning, and dense retrieval systems. My work centres on understanding how neural architectures form abstractions, generalise under distribution shift, and encode predictive structure. I have independently designed transformer-based retrieval systems, explored embedding geometry through Matryoshka Representation Learning (MRL), and investigated hybrid architectures combining transformer representations with classical learning methods. These projects have increasingly pushed my interests toward architectural questions surrounding compositionality, predictive representations, and world-model learning. Alongside this work, I filed a published patent (2025) for a lightweight IoT encryption algorithm, reflecting my ability to develop technically novel systems and carry ideas through formal research and publication pipelines. I aim to contribute long term to research in representation learning, transformer architectures, and predictive world models.

EDUCATION

Integrated Master of Computer Applications (MCA) *Aug 2022 – May 2027 (expected)*

Amrita Vishwa Vidyapeetham, Mysuru

- Coursework: neural network mathematics, optimisation theory, and deep learning architecture design.
- Independent research in transformer architectures and information retrieval (see Projects below).

HONOURS & EXAM SCORES

GATE 2026 – Valid Score *2026*

- Qualified the Graduate Aptitude Test in Engineering (GATE) 2026 with a valid score, demonstrating strong fundamentals in computer science and engineering.

Second Place – Mind Meld Debate Competition *Nov 2024*

- Amrita Kalaanveshane inter-collegiate cultural fest.

EXPERIENCE

Junior AI Engineer (Part-time) • KozkerTech • Kerala, India (Remote) *Jun 2026 – Present*

- Continuing as a part-time Junior AI Engineer following the successful completion of the internship, taking on expanded ownership of AI framework development.
- Leading ongoing improvements to the OpenClaw-based CRM pipeline, refining automated data flows connecting customer touchpoints and reducing manual intervention in lead and contact management.
- Collaborating with the engineering team to architect new agentic workflow components for internal business process automation.

AI Engineer Intern • KozkerTech • Kerala, India (Remote) *Feb 2026 – Jun 2026*

- Built and developed AI frameworks to automate internal and external business workflows.

- Played a key role in developing a CRM pipeline using OpenClaw, building automated data flows that connected customer touchpoints and reduced manual intervention in lead and contact management.

Machine Learning Researcher • STM Document Engineering Pvt Ltd • Thiruvananthapuram, Kerala (Remote) *Jun 2025 – Jul 2025*

- Developed ML-powered tools to aid in polishing and post-processing typeset texts within the LaTeX workflow, contributing to STM's typesetting solutions pipeline.
- Built subsidiary automation tools to streamline the LaTeX authoring and review workflow, applying Python and machine learning to a document engineering context.

RESEARCH PROJECTS

ForgeLang — Agent Programming Environment • Personal Project *2025 – Present*

- Designed and implemented a full-stack agent programming environment centred on a purpose-built declarative language (FORGE) for expressing agent goals, memory, constraints, tools, and multi-agent workflows.
- Built a six-module platform: a syntax-aware FORGE IDE with live agent inspection, a visual node-canvas builder (React Flow), an async execution studio with real-time reasoning traces, a monitoring observatory (Recharts), and a community marketplace for sharing agent architectures.
- Architected a contract-first monorepo (pnpm workspaces) with an OpenAPI spec as single source of truth, auto-generating React Query hooks and Zod schemas via Orval; backend on Express 5 + PostgreSQL + Drizzle ORM.

Cosmogony — Procedural World Forge & Text RPG • Personal Project *2025 – Present*

- Built a seeded procedural world-generation engine and space-fantasy text RPG, deployed to production on Vercel; players configure universe parameters (threat ratios, magic/tech balance, custom lore) to generate unique coordinate-mapped worlds.
- Implemented an SVG constellation vector map with node-to-node travel, atmospheric hazard mechanics, faction reputation blockades, and a keyword-driven narrative console supporting natural-language commands.
- Engineered a dynamic RPG attribute system (VIT, STR, AGI, INT) scaled from generative seed parameters, with procedural chronicle origins, session-persistent authentication, and a real-time Explorer Soul Panel overlay.

Cross-Lingual Information Retrieval (Hinglish / Manglish) • Independent Research *2025*

- Built a hybrid retrieval pipeline for code-mixed Hinglish and Manglish queries over English corpora (Banking77), combining custom BM25 with KLD alignment and domain-specific query expansion against a fine-tuned SBERT bi-encoder.
- Integrated Matryoshka Representation Learning (MRL) and a FAISS vector index; applied Reciprocal Rank Fusion (RRF) to merge lexical and semantic scores, achieving MRR ~0.903 on the hybrid configuration.
- Added cross-encoder re-ranking (ms-marco-MiniLM-L-6-v2) and comprehensive evaluation across MRR, MAP, P@k, and R@k; extracted hard negatives for iterative fine-tuning.

Chrahya Series Alpha — Zero-Knowledge Cognitive Neural Agent • Independent Research *2025 – Present*

- Designed and implemented Chrahya, a text-based neural agent operating under the Consciousness Instrumentality Protocol (CIP), initialised with randomised parameters and trained entirely online from live user text, webcam video, and microphone audio.

- Implemented continuous background consolidation with a FastAPI/PyTorch server exposing LEARN, CONSOLIDATE, ENGAGE, and DISENGAGE state transitions; model weights persist to disk across sessions (state/memory.pt).
- Documented full architecture, operational state machine, and a serialisation/resizing test suite in the public alpha release.

Sugarcane Leaf Disease Detection: ViT + SVM Hybrid • Collaborative Faculty Research • Amrita Vishwa Vidyapeetham *Nov 2024 – Apr 2025*

- Developed a hybrid deep learning pipeline combining Vision Transformer (ViT) feature extraction with an RBF-kernel SVM classifier, achieving 96% test accuracy.
- Conducted comparative analysis against KNN, Decision Tree, and Random Forest baselines to study how transformer-derived representations alter downstream decision boundaries.
- Observed that transformer feature representations reduced intra-class variance and improved separability beyond what classical classifiers achieved independently.

Grammar Correction Tool • Personal Project *Jun – Jul 2025*

- Built a fully offline, modular grammar-correction system combining rule-based logic with ML components, designed as an extensible NLP research platform for academic and domain-specific writing.

Smart Traffic Congestion Visualiser (STCV) • Personal Project *Sep 2025 – Present*

- Built a lane-scheduling simulator using a custom starvation-aware priority heuristic (Priority = Vehicles + Starvation_Time × 0.25), preventing lane starvation under varying congestion conditions.

Stock-War Correlation Analysis • Data Analysis *Nov 2024*

- Analysed arms, petroleum, and gold stock trends during geopolitical events via news APIs and sentiment analysis; identified media-bias limitations in sentiment-driven prediction, informing future work on robust signal extraction.

PATENT

NRG Morph: Lightweight IoT Encryption Algorithm *Filed May 3, 2025*

Application No. 202541043088 • Status: Published • Amrita Vishwa Vidyapeetham

- Invented a resource-constrained encryption algorithm for real-time smart-grid IoT devices, balancing cryptographic integrity with low compute overhead on embedded hardware.
- Demonstrates ability to develop novel, IP-worthy technical solutions and carry research through formal publication pipelines.

TECHNICAL SKILLS

ML / DL Frameworks: PyTorch, HuggingFace Transformers, scikit-learn, FAISS, sentence-transformers, FastAPI

Architectures & Methods: Transformers (ViT, BERT-family), bi-encoders, cross-encoders, MRL, SVMs, GANs, RAG, multi-agent systems

Languages: Python (primary), TypeScript, C/C++, Java, C#, Bash, Lua

Frontend & Full-Stack: React 19, Vite, Tailwind CSS, React Flow, Recharts, Express 5, PostgreSQL, Drizzle ORM

Tools & Platforms: Linux (Arch), Neovim, Git, pnpm workspaces, Docker, LuaLaTeX

COURSES & CERTIFICATIONS

- Reinforcement Learning, Infosys Springboard (Oct 2023): Fundamentals of RL and its mathematical framework.
- Generative AI Workshop, Amrita IIC Research Lab (Sep 2024): Embedding models, self-attention; implemented a RAG chatbot.
- Foundations of Cybersecurity, Google (Nov 2025): Part of the Google Cybersecurity Professional Certificate.

CONFERENCES & PROFESSIONAL MEMBERSHIPS

- TUG 2025 (TeX Users Group Conference), Thiruvananthapuram, Jul 2025: Engaged with advances in LaTeX tooling, tagpdf for accessible PDF generation, and CTAN ecosystem maintenance.
- Member, TeX Users Group, Jul 2025 – Present.

LANGUAGES

Malayalam (Native) • **English** (C2 Proficient) • **French** (A1 Beginner)